

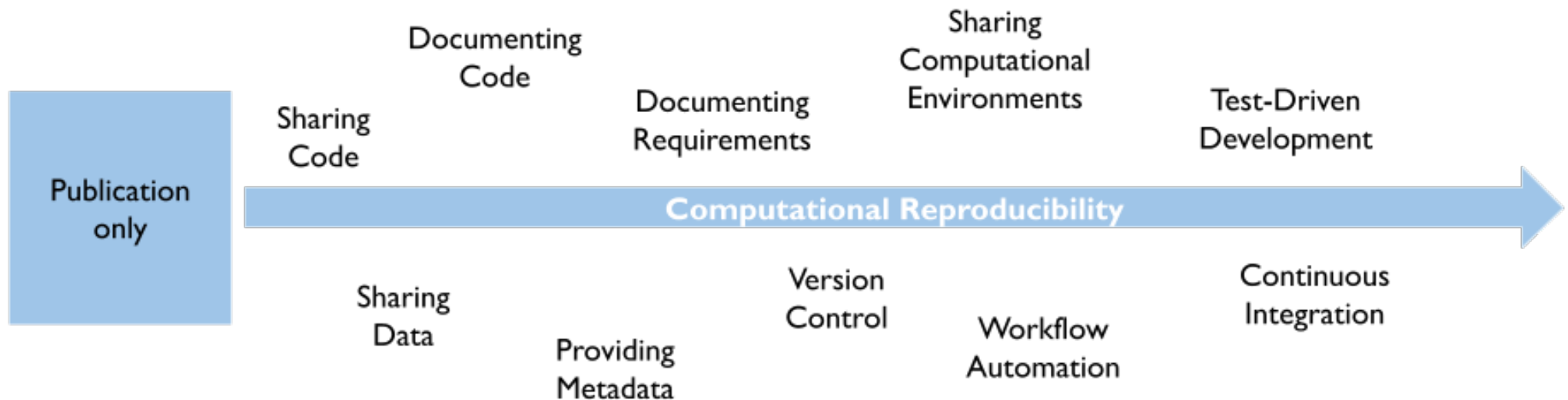
Advanced Workflows

Combining dynamic reporting with version control, automation, and containerization

Samuel Pawel, Peter Degen, Fabio Molo, Gorka Fraga González

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Reproducibility is a spectrum



- Goal today: Understanding **what is possible** towards the right side
- ~~Goal today: Deep technical understanding of advanced workflows~~

Dynamic reporting solves copy-paste problem, but ...

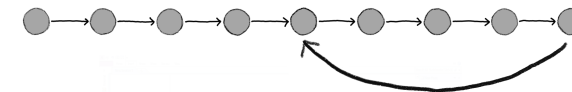
- How to keep track of different **file versions**?
 - `paper-final.qmd` → `paper-final2.qmd` → `paper-final2b.qmd` → 😱
 - Solution: **Git**
- How to know which files to run in which **order**?
 - `functions.qmd`, `analysis.qmd`, `plots.qmd`, `paper.qmd`, 🤔
 - Solution: **Make**
- How to make sure code still runs when **software versions** change?
 - **Today**: R version 4.5, ggplot2 3.5.2 → **Next year**: R version **?**, ggplot2 **?**
 - Solution: **Docker**
- Very nice paper on such a workflow: Peikert and Brandmaier ([2021](#))

Version control with Git

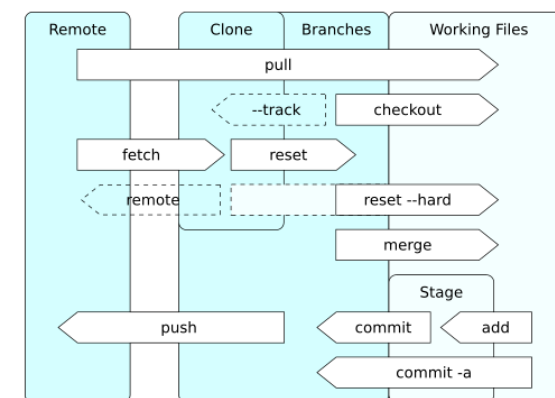
- De facto standard version control system
- Can go back in time to previous versions
- Can track changes between versions
- Can branch and merge
- Useful for collaboration and file transfer
- Command line interface (learning curve)
- Graphical interface in many editors/IDEs (e.g., RStudio, VS Code)
- Git (software) ≠ GitHub/GitLab (Git repository hosting services)
- Snapshots enable persistently archiving (with DOI) on data repositories (e.g., Zenodo)



<https://git-scm.com/downloads/logos>



The Turing Way Community and Scriberia (2024)



[https://commons.wikimedia.org/wiki/](https://commons.wikimedia.org/wiki/File:Git_operations.svg)

Automation with Make

- Tool for managing **computational recipes**
- Automating multiple steps into one

Arrabbiata sauce, or sugo all'arrabbiata in Italian, is a spicy sauce for pasta made from garlic, tomatoes, and dried red chili peppers cooked in olive oil.

https://en.wikipedia.org/wiki/Arrabbiata_sauce

Makefile

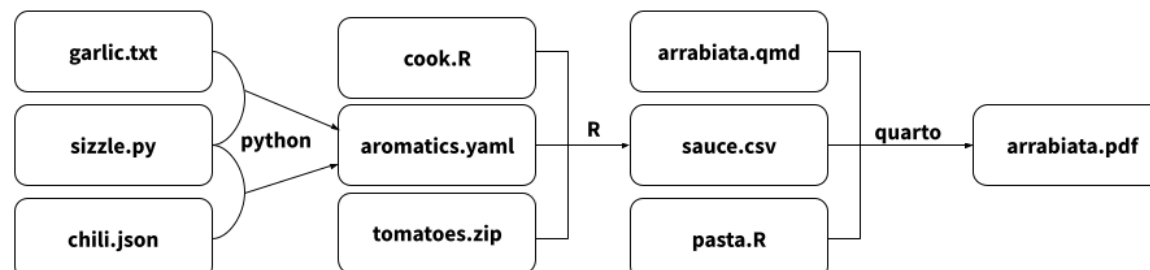
```
arrabiata.pdf: arrabiata.qmd sauce.csv pasta.R
    quarto render arrabiata.qmd

sauce.csv: cook.R tomatoes.zip aromatics.yaml
    R -e "source(cook.R) "

aromatics.yaml: sizzle.py garlic.txt chili.json
    python sizzle.py garlic.txt
    python sizzle.py chili.json
```

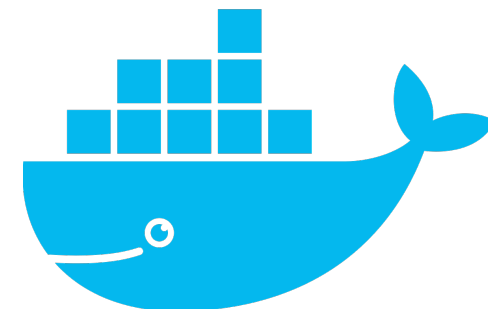
Terminal

```
make arrabiata.pdf # execute recipe
```



Containerization with Docker

- **Software containers** encapsulate computational environment
- Computational environment components:
 - Operating system
 - R and R packages
 - Numerical software libraries (e.g., LAPACK, BLAS)
 - other software (e.g., LaTeX, Java)
- **Docker** is a popular tool for containerization
- **Rocker** is a Docker image for R (<https://rocker-project.org/>)



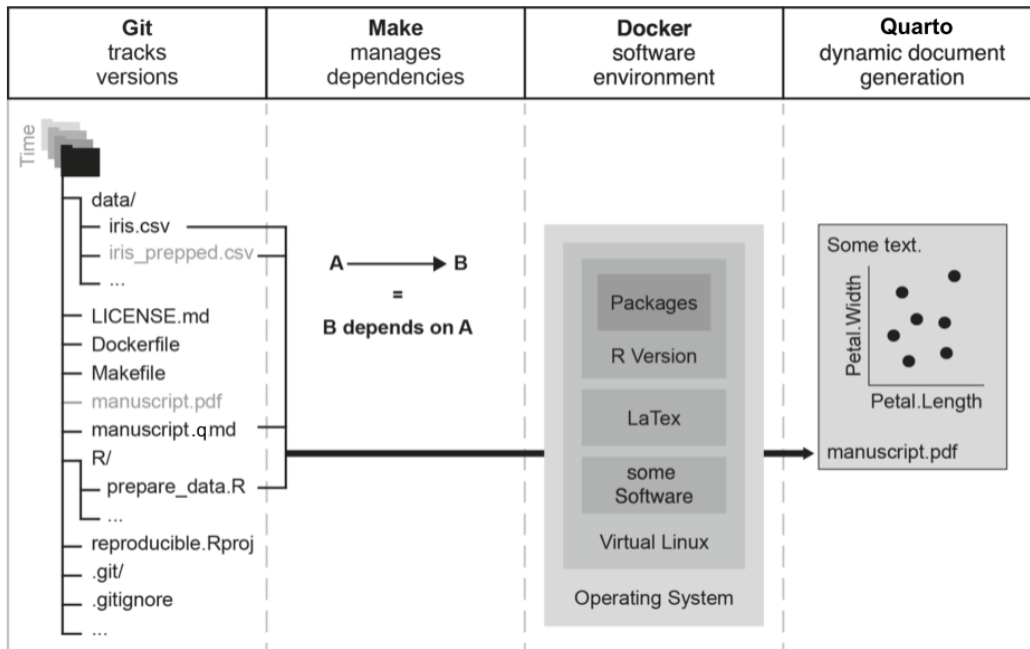
[https://commons.wikimedia.org/wiki/
File:Docker-svgrepo-com.svg](https://commons.wikimedia.org/wiki/File:Docker-svgrepo-com.svg)

Dockerfile

```
# Uses a Rocker container image with R version 4.4
FROM rocker/verse:4.4

# Install R package versions at time when R 4.4 was up-to-date
RUN install2.r ggplot2 dplyr kableExtra
```

Putting it all together



Terminal

```
# clone git repository
git clone git@gitlab.uzh.ch:crsuzh/zurich-names.g

# enter project repository
cd zurich-names

# run automated analyses
make docker-report
```

→ **report.html**

<https://gitlab.uzh.ch/crsuzh/zurich-names>

Modified from Peikert and Brandmaier (2021)

Conclusions

- Dynamic reporting + version control + automation + containerization
 - improve **reproducibility, reuseability, collaboration**
 - lends research more **transparency, credibility**
- There is a **learning curve** 🤔, but ...
 - start simple and learn **step-by-step** 💡
 - tools are **constantly evolving** 🚀
 - invested learning time can **save you time** in the future ⌚
 - skills are **useful in/outside Academia** 💰🧪

The Swiss Reproducibility Network Academy

- **Early career** researchers section of the SwissRN
- Goal 1: Connect young researchers interested in **reproducibility** in **Switzerland**
- Goal 2: Improve reproducibility of research in Switzerland
- How to join?
 - young researchers in **Switzerland** (working language: English)
 - **interest** in improving research and reproducibility
 - **every field** is welcome (diversity is the key!)
- More information: <https://www.swissrn.org/contents/academy/>



CRS Primers

Primer: Managing Complex Data Analysis Workflows

[Peter Degen](#), [Christian Meesters](#), [Gorka Fraga-González](#), [Samuel Pawel](#), [Fabio Molo](#)

doi: [10.5281/zenodo.15789797](https://doi.org/10.5281/zenodo.15789797)



Motivation

A typical data analysis workflow (Fig. 1) consists of input files (data, configuration files) that are processed with various tools (software packages, custom scripts) to generate output files (summary tables, figures, or even an entire manuscript). If the workflow is simple, one might be tempted to manage this workflow “manually” (e.g., individually calling scripts from the command line) or by writing a short Bash script that sequentially performs all necessary tasks.

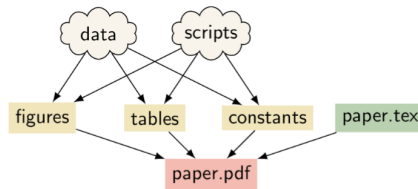


Figure 1: Example data analysis workflow. Figure sourced from [The Turing Way](#) handbook under a CC-BY license.

<https://doi.org/10.5281/zenodo.15792190>

Primer: Software Containers for Reproducible Research

[Gorka Fraga González](#), [Francesco Garassino](#), [Fabio Molo](#), [Steffen Bollmann](#), [Eva Furrer](#)

doi: [10.5281/zenodo.13757230](https://doi.org/10.5281/zenodo.13757230)



Why containers enhance reproducibility

Sharing analytic code along with open data is often not sufficient to ensure the computational reproducibility¹ of research results. Computational results depend not only on the analytic software and third-party packages used, but also on additional software and underlying system components. When any of these dependencies receive updates or become unavailable, there is a risk that the analytic code cannot be executed anymore or produces results that differ from the original results.

Software containers can be a solution to this problem, as they provide a way to create a snapshot of the entire computing environment used in an analysis, including all system dependencies. Some use cases are re-running an old analysis, scaling up an existing analysis (e.g., increasing the sample size), running code in different operating systems or sharing an analysis pipeline including software and data. This primer introduces containerization concepts, workflows and platforms so that researchers with limited IT expertise or no access to dedicated staff can get started with improving the reproducibility of their computational analyses.

<https://doi.org/10.5281/zenodo.13757230>

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